Platform observers will therefore conclude that Itchy set his clock to 12:00 *before* Scratchy and will therefore claim that Itchy's clock is set a bit ahead of Scratchy's. For example, to a platform observer like Martin, when it's 12:06 on Itchy's clock, it may be only 12:04 on Scratchy's (the precise numbers depend on the length and the speed of the train; the longer and faster it is, the greater the discrepancy). Yet, from the viewpoint of Apu and everyone on the train, Itchy and Scratchy performed the synchronization perfectly. Again, although it's hard to accept at a gut level, there is no paradox here: *observers in relative motion do not agree on simultaneity—they do not agree on what things happen at the same time.*

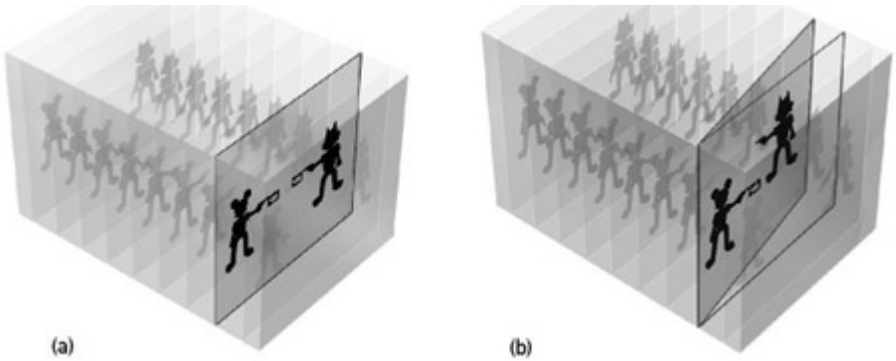


Figure 3.4 Time slicings according to **(**a**)** Apu and **(**b **)** Martin, who are in relative motion. Their slices differ by a rotation through space and time. According to Apu, who is on the train, the duel is fair; according to Martin, who is on the platform, it isn't. Both views are equally valid. In **(**b**),** the different angle of their slices through spacetime is emphasized.

This means that one page in the flip book as seen from the perspective of those on the train, a page containing events they consider simultaneous—such as Itchy's and Scratchy's setting their clocks—contains events that lie on *different* pages from the perspective of those observing from the platform (according to platform observers, Itchy set his clock *before* Scratchy, so these two events are on different pages from the platform observer's perspective). And there we have it. A single page from the perspective of those on the train contains events that lie on earlier and later pages of a platform observer. This is why Martin's and Apu's slices in Figure 3.4 are rotated relative to each other: what is a single time slice, from one perspective, cuts across many time slices, from the other perspective.